

SOLID STATE INDUCTION HEATING EQUIPMENT

OPERATION AND MAINTENANCE INSTRUCTIONS

HOTSHOT

3.5, 5, 7.5kW POWER SUPPLIES

File Name:
Part Number:
Revision:
Date:
By:
Author:

001 0282k close
001 0282
K
0707 00
Wend

ALLAN PETERSON

Derivation:

This K-revision has been written to

- reflect changes to RHS tapping method, hardware

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Notes:
In this manual, these symbols indicate:

- ✓ a step in a procedure
- § manual sections
- ★ lighted indicators on the equipment





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1 Introduction

Your HOTSHOT™ is a solid-state induction heating system that converts three-phase line voltage to a 3.5, 5, or 7.5kW output (depending on model) over a range of radio frequencies (RF) and voltages. This energy is delivered to a remote series-resonant circuit - including your coil - where a precisely controlled magnetic field is created around your work-piece.

HOTSHOT™ uses a special sensing technology enabling greater control of your process, precise frequency registration under varying load conditions results in increased precision in heat control and times-to-temp.

Special features

Your HOTSHOT simplifies and improves your heating results with:

- Touch-pad interface for control and programming
- Backlit LCD simultaneous display of up to four data fields
- electrically isolated remote heat station

The power from your supply can be controlled:

- manually from the front panel
- remotely by your signals provided to the rear panel
- remotely through a configurable serial port
- from one of four 5-step profiles you specify

This selection is made from the touch-pad and can be changed at any time

Unpacking

Your Ameritherm HOTSHOT Induction Heating system has been carefully packed to arrive at your facility in good condition.

We suggest you inspect the shipping carton in the presence of the carrier when the unit arrives at your facility. Look for dents, crushed corners or torn cartons.

YOU MUST REPORT ALL DAMAGE DIRECTLY TO THE CARRIER.

✓ Check to see that all the parts we shipped arrived at your plant:

- HOTSHOT power supply
- remote heat station with cable
- reference coil, attached to mounting blocks
- this manual



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If you note damage, do not discard any of the exterior packing cartons; retain the carton(s) so that the carrier's claim agent may inspect it. Ameritherm will assist you in filing a claim against the carrier should it be necessary to do so.

1.1 Safety Considerations

Your HOTSHOT uses RF energy to raise the temperature of your work-piece. Most of this RF energy is transmitted into the work-piece and some is transmitted into the air.

Physical Risks

During the ★HeatOn cycle, high RF voltages are present at your work-coil. There are several risks, which we instruct you to protect against:

- **High Temperatures:** take steps to prevent personal contact with the work-piece you are heating. Severe burns can result from contact at these temperatures!
- **RF Voltages:** we recommend that you provide protection against personal contact with the work-coil when it is energized (★HeatOn).
- **Induction burns:** the energized work-coil causes nearby metals to become heated; **DO NOT WEAR JEWELRY OR CARRY KEYS** within a few inches or centimeters of the energized work-coil.



Safety Circuits

Your optional normally-closed E-stop switch can be installed in series with the system 24Vdc supply (CTB1:15-16); when the switch opens, all output power and operations cease. When the switch is re-closed, the power supply returns to ★Ready (non-hazardous state). Your E-stop switch must be isolated (dry-contact), capable of switching 3A @ 24Vdc.

The power supply is fitted with an external rear panel jumper (CTB1:15-16) which must be removed to insert any safety switches required.

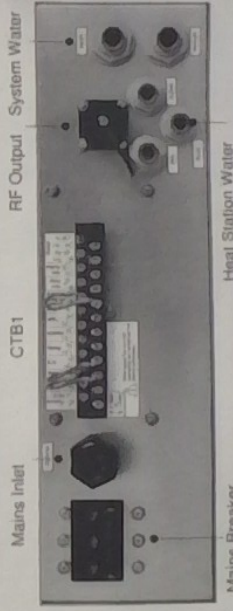
1.2 Front and Rear Panels



Touch-pad & Display

All use-interface is conducted here

- Status LEDs indicate what your HOTSHOT™ is doing
- AC indicator is lighted when unit is turned on
- LCD display provides operating information, navigation/programming prompts
- START/STOP buttons enable front panel control of heating
- navigation buttons are used to program and control the unit

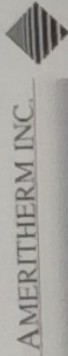


Mains Breaker

This gang switch controls the mains connection to the power supply. Trip level may vary by model, see §4

Mains Inlet

This strain-relief fitting secures your mains cabling



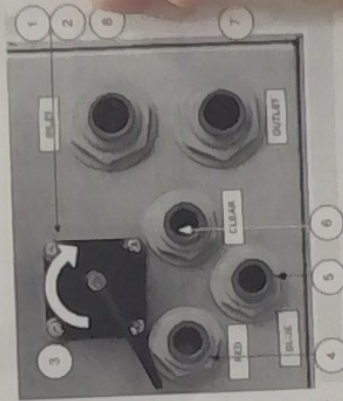
1.4 Assembly

Take these factors into consideration as you determine where to install your system:

- availability of water and drain
- availability of 3Ø AC power

Attach heat station & water

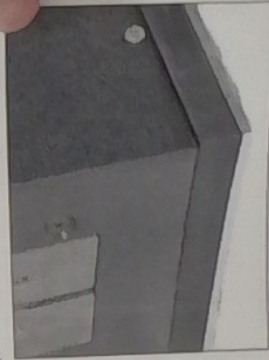
- ✓ ① unscrew protective cap
 - ✓ ② insert indexed heat station plug
 - ✓ ③ tighten ring out
 - ✓ ④ insert RED tube from heat station
 - ✓ ⑤ insert BLUE tube from heat station
 - ✓ ⑥ insert CLEAR tube from heat station
 - ✓ ⑦ insert tubing for water outlet fully into bottom port
 - ✓ ⑧ insert tubing for water inlet fully into top port
- to remove tubing
- ✓ push in on retaining rings, pull out tubes



Heat station mounting base options

Your heat station has been fitted with a two-part mounting insert, which elevates and insulates the bottom of the unit from your metal tabletop or mounting fixture, reducing the amount of secondary heating you might experience under certain sustained operating conditions. Secured with four M6 nylon screws, this base can be utilized or omitted in your mounting scheme. To mount:

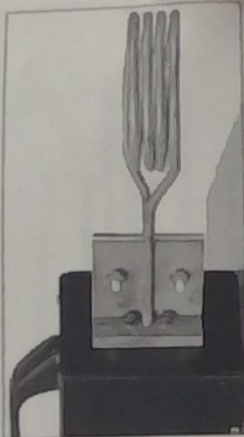
- ✓ remove nylon screws, use four M6 stainless steel screws. For length, add 1" to your fixture depth to include plate, add 3/8" to mount without plate.



Work-coil or reference coil (included with the unit)
A reference coil is included with your heat station to assure basic operation; while it is suitable for use in heating, another coil may have been ordered for your use.

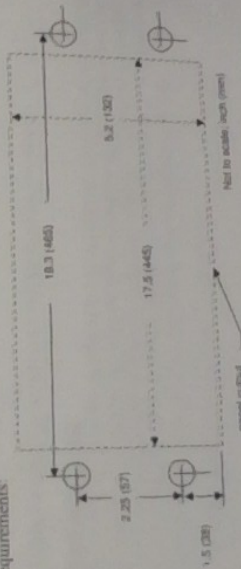
If you are attaching your own coil:

- ✓ use 4mm Allen-wrench to remove the 4 screws securing the reference coil
- ✓ remove reference coil and retain 2 O-rings from coil mounting plate
- ✓ re-use or use new O-rings; insert in recessed face of heat station
- ✓ secure coil with 4 screws provided; tighten



Rack/panel Mounting

Rack/panel Mounting
Your HOTSHOT is mountable in your panel or 19" rack (3U); observe these steps and requirements:



- ✓ unscrew/remove 4 rubber feet on unit base
- ✓ (panels) provide cutout and mounting pattern per above; plan for M5 or M6 screws
- ✓ ensure that rear of unit is supported when mounted
- ✓ provide at least 24" depth to permit water and AC cabling bend
- ✓ secure unit with M5 or M6 hardware



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Cooling water

In the process of heating your work-piece, the current flowing in the heat station and power supply heats components, too. HOTSHOT™ provides a means of directing fresh water into and through the components, enabling the water to pick-up the excess heat and transfer it, either to drain or to cooling to be recycled back to the equipment.

✓ Follow these important restrictions:

- If used, fill re-circulating system with distilled water
- Do not use de-ionizing water systems
- Do not use inhibited glycols as anti-rust or anti-freeze agents
- Do not use inhibitors supplied by Ameritherm. Install a pressure regulator and a particulate matter filter (both non-ferrous) in the supply of your cooling water (If cooler is not supplied by Ameritherm)
- Do not use ferrous pipe; it can lead to serious equipment damage!
- ✓ Establish a cooling water source capable of 40psi (minimum) differential pressure across the system and at least 1.5 gpm (5.7 l/min) flow
- ✓ Provide inlet and outlet water through 1/2" OD plastic or copper tubing
- ✓ Cut each tube end cleanly and remove any burrs
- ✓ Insert drain tube end fully into lower fixture labeled OUTLET (", above)
- ✓ Insert supply tube end fully into upper fixture labeled INLET (", above)

(To remove tubing, press retaining ring while pulling on tube.)

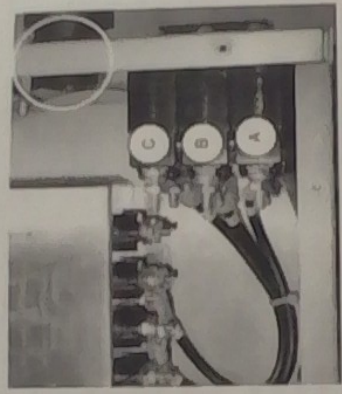
Flow monitoring

The amount of water flowing through your equipment is monitored, since operation with inadequate or absent cooling water can result in damage to the coil or capacitor. When flow is adequate to the heat sink and to the coil - see §4 - two internal switches are closed. When flow falls off to the heat sink or the coil, one or both of the switches open and the power supply senses it, halts RF power and indicates ★ Fault

✓ No action required by you, internal connections.

Mains connection

- Your HOTSHOT[®] power supply requires 3Ø AC power and ground; you provide this service with your cable at the rear of the power supply:
- ✓ use Phillips screwdriver, remove 12 cover screws
 - ✓ remove plastic safety shield by removing screw at center
 - ✓ select wire gauge from §4.2 current data
 - ✓ remove 4" cable jacket; strip 4 wires for crimp ring-lugs
 - ✓ insert ~12" of your 4-conductor cable through strain-relief fitting at rear (circle); this accepts cables from 0.51 - 0.71 in (13 - 18mm); tighten ring to secure cable
 - ✓ crimp 4 ring-lugs to wires
 - ✓ use nut driver, remove nut-washer combinations from circuit breaker line-side terminals A - B - C
 - ✓ attach line wires to A-B-C; replace nut-washer combinations; tighten
 - ✓ attach ground wire to terminal marked with safety ground symbol
 - ✓ replace safety shield; secure center screw
 - ✓ replace unit cover; replace 12 screws



Start-up display

Upon power-up, certain important information about your HOTSHOT is displayed:

Output power capability: Depending on the unit you purchased.	HotShot 7.5 7.5kW RF Power
Firmware versions: There may be instances where we need this information from you. n.nn represents the version numbers of the Display or Main firmware	Display: vn.nn Main: vn.nn
System ID# This number is unique to your unit. You will need this number if you elect to purchase any of a group of feature unlock codes from Ameritherm	System ID #: 12345P
You can also view these screens when the unit is ON; while in the Control Zone, §2.2, press ▲ or ▼ to repeat the Start-up display screens	

The next section explains what you see when the unit is operating in the Home screen.



2 How Your HOTSHOT Works

When you turn **HeatOn**, (details, next page) this equipment concentrates an electromagnetic field within the coil diameter. When you put your metal part into the coil, this field causes internal electric currents in your part. Friction from these currents heats your part.

With **HeatOn**, both Setpoint (your selected output level — higher means hotter) and Output Level (actual power or current delivered to your part) are displayed. Typically, the Output Level is highest when your part is in the coil and it returns to a lower level when you remove your part. (For detailed explanations or to understand operation that is not typical, see §2.3.)



LED Status and system notices are relevant if your part is in the coil during **HeatOn**. Status and notices displayed after you remove your part can be ignored.

This equipment is designed to operate simply, requiring you to have very little knowledge of induction. This section presents HOTSHOT operation in two parts:

- **Home zone:** your HOTSHOT operates in this zone when you first apply power. The factory settings enable you to get right to work heating parts by operating the front panel. From the Home zone, you can adjust output levels, set the timer and turn RF power on and off. Simple heating jobs may never require you to use features outside this zone.
- **Control zone:** this is how you make use of the full set of HOTSHOT capabilities by changing any of the operating characteristics. Refer to this section to customize your HOTSHOT.

Next is a simple example of operation from the Home zone. In it you will operate your HOTSHOT by:

- starting from the front panel
- using 2-button **START** (requires **STOP** to turn off heat)

(Other operating characteristics are discussed in the Control zone section.)

After this example, the front panel display, LED and touch-pad features are explained. The Control zone is explained in detail in the next section.

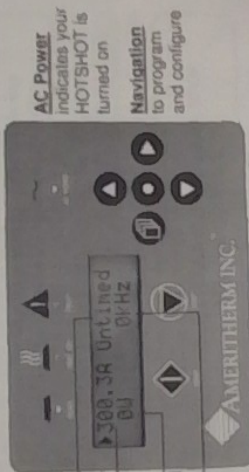
2.1 Home Zone

When you turn it on, HOTSHOT operates in the **Home zone** (illustration). Within this zone, you can adjust output levels, set the timer and turn RF power on and off.

Home Zone Display

Status LEDs

Indicate what your HOTSHOT is doing:
 READY: waiting to heat; no faults detected
 HEAT ON: power is being output
 FAULT: heating interrupted, fault detected

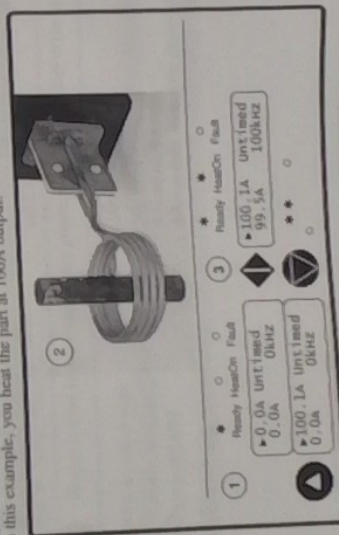


Timer cycle or continuous Tap adjustments are made with 3 wires (X, Y, G) and a jumper (Z), which are attached as needed in a 3x3 terminal block.

START/STOP
to initiate heating from front panel

Simplest example...

In this example, you heat the part at 100A output.



2) How Your HOTSHOT Works

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Do this...	Observe...
4 open water valve for the system 4 check the work coil 4 check condition of (any) safety barriers or limit switches 4 turn on HOTSHOT™ 4 ① note display 4 press ▲ adjust until	check drain, assure water is flowing assure there is no work-piece yet all safety devices are clear, electrically closed ★ AC Power and ★ Ready LEDs assure Setpoint is selected (example) 100A
4 ② secure unheated part in coil 4 ③ press START 4 press STOP	Within the Home zone, use these buttons to adjust Setpoint and Timer (indicated with the display pointer ► 1; press once to make a single-step adjustment, or you can hold the button to scroll quickly to a new value. part not contacting coil, do not hold ★ HeatOn, ★ Ready ★ Ready only
4 ④ Press START to begin induction heating (Home zone only) Press STOP to halt induction heating (2-button start type, below) STOP signal from front or rear panel always halts induction heating	avoid contact with heated portions
Timed heating... In this example, you heat the part at 100A output for 10 seconds	
Do this... 4 press ► select Timer	Observe... Untimed
4 ⑤ Button is used to move the display pointer from one data field to the next 4 press ▲ adjust until: 4 ⑥ secure unheated part in coil 4 ⑦ press START 4 remove heated part from coil	10.00s part not contacting coil, do not hold ★ HeatOn, ★ Ready ★ Ready only, when timer expires (HeatOff) avoid contact with heated portions

Display Features

setpoint
Output

Timer
Freq

At the top left corner of the screen, this is your desired output level, displayed in RF amps. Larger values result in faster heating or hotter parts. With the pointer (►) on setpoint, you can increase or decrease the setting with the ▲/▼ buttons.

HOTSHOT works to match the RF output to this setting when the RF power is on (HeatOn). In the Home Zone, ► moves the cursor between Setpoint and Timer

setpoint
Output

Timer
Freq

At top right, use the timer to control the duration of your heating. With the pointer (►) on Timer, you can increase or decrease this setting to values between 30 milliseconds and 9999 seconds with the ▲/▼ buttons. The timer begins counting down when you START. To display elapsed cycle time, move the pointer to Unfired, then press STOP. Elapsed cycle time is then displayed during HeatOn. (Assure 2-button Start §2.2 when using timer.)

► to quickly reset the Timer to 0.00 if indicated (►)

setpoint
Output

Timer
Freq

At the bottom left corner of the screen, this is the actual RF output (always 0.0 when the heat is turned off). When the output power is turned on, this displays the output in terms of RF coil power or current. The output level can be shown in RF Amps or RF Watts (press Ⓢ to change).

setpoint
Output

Timer
Freq

At the bottom right corner of the screen, the frequency at which the supply is heating is shown. This is the resonant operating frequency and is the result of your coil in combination with the heat station's internal capacitance. (You cannot 'set' this frequency other than through your selection of heating coil.)

When ► Ready after first turning on the HOTSHOT, the frequency is displayed as 0 MHz. During the first heat cycle (► HeatOn), the frequency value shows you what the resonant (most efficient heating) point is. When the heat is turned off, the display will show you where it will begin to resonate on the next heat cycle based on the recent heating cycle. Sometimes, you may see that the output frequency approaches either end of the operation range of HOTSHOT and the display advises you to increase or decrease capacitance. Please call Ameritherm for assistance should this happen and we will supply you with the appropriate capacitors.

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Status LEDs

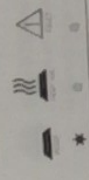
This guide will help you understand the operation of your HOTSHOT™.

LED Status and system notices are relevant if your part is in the coil during

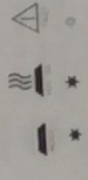
• **HeatOn.** Status and notices displayed after you remove your part can be ignored



Ready:
HOTSHOT is waiting to heat; no faults detected



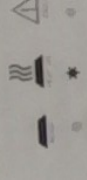
Heating:
START command received, outputting power per settings. It may be possible to achieve a more efficient configuration, though output is within acceptable margins.



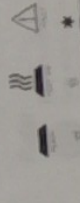
for possible suggestion to improve operation, refer to §3 for further adjusting instructions

Limit:

Unit heating, not meeting setpoint
• @display; output differs from setpoint by more than 2%
L • @display; pre-set Alarm Limit has been exceeded; for suggested corrective step(s)
... W (W overscored); displayed output power is outside a preferred tolerance band. Increase tap for possible improvements
!Cap @display; you are attempting to adjust Setpoint above safe limit for capacitors used. Reduce Setpoint



Fault:
Heating interrupted, fault detected, correction required
to Control zone. ▶ Status then for fault notice



★ AC Power; indicates HOTSHOT is ON; * see the Status screen for notices

Touch-panel Features

START/STOP

When starting the unit from the front panel, use these buttons to (de)activate heating.
Press **START** to begin induction heating. ★HeatOn (LED) shines and the display is updated with operating data. **START** operates only within **HOME** zone.

Press **STOP** to halt induction heating (2-button start type, below). The unit returns to ★Ready state. (STOP signal from front or rear panel always halts induction heating)

Next

Use it to move the display pointer (▶) from one data field to the next. For example, the display first looks like this:

▶0.0A Untimed
0.0A ★0kHz

To program a heating time, press ▶ once; the pointer is to the left of the 'Untimed' message. Any following adjustments will apply to the timer.

0.0A ▶Untimed
0.0A ★0kHz

Up/Down

Within the **Home** zone, these are used to adjust Setpoint and Timer (indicated with the display pointer). Within the **Control** zone (below), use either button to 'replay' the start-up screens (to review firmware version), to adjust alarm and RF-Set figures or to display Setup sub-fields.

To adjust Setpoint or Timer, you can press once to make a single-step adjustment, or you can hold the button to scroll quickly to a new value.

Page

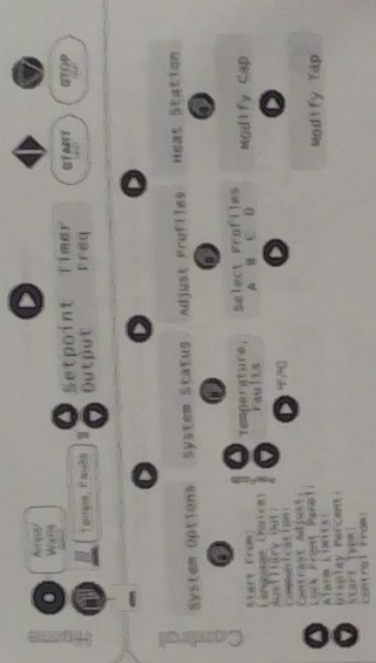
From within the **Home** zone, pressing this button moves to the **Control** zone (see map) if the equipment is ★Ready. Pressing page during ★HeatOn displays the internal temperatures and notices.

Within the **Control** zone, this button is used to select and exit the feature fields.

Home

This button takes you back to the **Home** zone, no matter where you have navigated in the touch-pad map.
If you are already at the **Home** zone, pressing this button switches between displayed output current (A) and displayed output Watts (W).

Navigation
 This map will help you understand operation and programming your HOTSHOT.
 Each touch pad button is represented in the map, showing the effect touching each one has depending on your location in the map.



Normal operation is from within the **Home** zone, where you:

- adjust setpoint and timer
- turn RF power on and off
- switch displayed units or
- move to the **Control** zone using

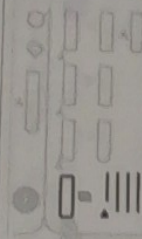
The **Control** zone is where you:

- display and select Setup options (front or rear control, etc., below)
- display Status (internal temperature, • Fault notices)
- construct and enact heating Profiles
- adjust for RF output stage configuration (tap settings, tank capacitor)

2.2 Control Zone

This is how you access or change operating characteristics of the HOTSHOT™ Control Zone is accessible only while heat is off.

From the Home area, press . The Control Zone menu is shown. Use  to select from 4 sub-menus, detailed below (defaults underlined>).



Menu Selection

System Options

Once this screen is displayed, press ; use  to select operating features; use  to select the selections starting with the bottom of the list, working up.

Control From

Front Panel: during • HeatOn, RF output follows sequence entered at front panel.

Rear Panel: your existing signal controls output.

File A...: use one of four profiles you specify (07-3).

RS485 4 wire: if • Communication = RS485 4 Wire, below, select this to adjust output with serial command.

Control From

Front Panel

Start Type

To change START type

2button: • HeatOn when START is pressed (or CTB1 signal is closed) and remains on when released. • HeatOn ceases when STOP is pressed or when timer expires.

1button: • HeatOn only while button is pressed (or CTB1 signal closed), ceases when button is released (or CTB1 signal opens).

Start Type

2 Button

Display Percent

To enable/disable use of % displaying output levels

Off: if not used

On: in use

Adjust: see §3.3 Alarm Limits for instructions

Display Percent

Off

2) How Your HOTSHOT Works

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- ▶ **Alarm Limits**
 To enable/change Alarm settings:
 Off: if not used
 On: in use
 Adjust:
 see §3.3. Alarm Limits for instructions

▶ **Lock Front Panel**
 To prevent changes to settings:
 NO
 Yes: disables all buttons except START STOP.
 select and follow instructions, enter 98889 code
 To unlock, press @ while heat is off, re-enter code

▶ **Contrast Adjust**
 To change the contrast of the displayed characters
 (lower number = higher characters)
 Use ▶ to start, then ▲/▼ to adjust, use ▶ to save
 setting. (Setting is retained.)

▶ **Communication**
 Naming
 RS485 4 Wire (only protocol at this writing)
 Use ▶ to select, then ▲ to enable (if RS485 4 Wire)
 See §3. Serial Communications

▶ **Auxiliary Out**
 Feature not supported at this time

▶ **Language Choice**
 English: select action for available language(s)

▶ **Start From**
 Front: to START heating from the front panel
 Rear: you attach your remote signaling source
 (PLC, relay, etc./to CTB) at the rear and
 START with those signals.
 RS485 4 wire: if "Communication = RS485 4
 Wire" below, select this to start heating with
 serial command (STOP always works from front
 or rear)

▶ Alarm Limits
▶ Off

▶ Lock Front Panel
▶ NO

▶ Contrast Adjust
▶ To change 47

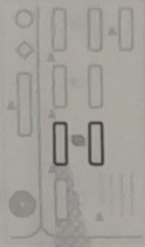
▶ Communication
▶ None

▶ Auxiliary Out
▶ Disabled

▶ Language Choice
▶ English

▶ Start From
▶ Front Panel

Menu Selection ►System Status



(Once this screen is displayed, press ; use /▼ to select operating features):

Display the temperature of active components on the heat sink (HS) and the internal ambient air (AM) and any active (unresolved) Fault notices (§2.3)

- ✓ use /▼ to display any previous Fault notices (previous = from last heat cycle)
- ✓ use to change between °F and °C

HS65 °F AM72 °F
Increase Tap

Menu Selection ►Adjust Profiles

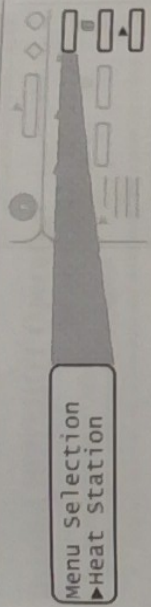


(Once this screen is displayed, press ; use /▼ to select operating features):

Adjust Profiles

This lets you build up to four independent sequenced-heating profiles (A-D); you set the starting and ending output levels and the duration for as many as five discrete steps per Profile (e.g., A1-A5, Multi-step, following section)

Select Profile
►A B C D



Menu Selection
▶Heat Station

(Once this screen is displayed, press ; use / to select operating features):

Heat Station

This enables the power supply to operate after you make changes to the heat station configuration, or if you select from two different heat stations. (Factory settings need never be changed if you make no changes to the heat station.)

To change, press for changes to either capacitor or to transformer tap settings

Although factory-set, you must 'inform' the HOTSHOT of any changes you make to the RF transformer taps or tank capacitor (see §3, tap and capacitor changing instructions)
✓ select the correct heat station model (/)

1000S
1000P
1000
800S
800P
400S
400P
300S
300P
150

Custom

(This information provided by Ameritherm)
If you have changed the transformer tap setting (§3.4), you must 'inform' the power supply
✓ select the correct tap setting (/)

Failure to change Tap or Cap settings following a change may result in false displays and less of component protection. This does not change capacitance.



Whenever you are unsure about the correct tap selection for your application, follow this rule: **start high, work down**. Beginning from the highest transformer taps presents the least stress to the HOTSHOT.

Choose Menu
▶Modify Tap

Choose Menu
▶Modify Cap

Choose Model
⇄150

to select capacitance

C1 value
⇄0.10uF

Select Tap
⇄18

2.3 Detailed HOTSHOT Operation

This section details how to:

- interpret the Status LEDs
- understand system faults and advisories
- program and execute continuous and multi-step heat-cycles (profiles)

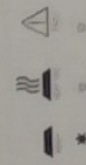
Interpret Status LEDs

This guide will help you understand the operation of your HOTSHOT.

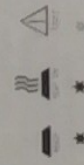


LED Status and system notices are relevant if your **part is in the coil during**

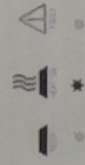
• **HeatOn**. Status and notices displayed after you remove your part can be ignored



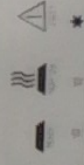
Ready:
HOTSHOT is waiting to heat; no faults detected



Heating:
START command received, outputting power per settings. It may be possible to achieve a more efficient configuration, though output is within acceptable margins.
[Icon] for possible suggestion to improve operation, refer to §3 for further adjusting instructions



Limit:
Unit heating, not meeting setpoint
• [Icon] @display; output differs from setpoint by more than 2%
• [Icon] @display; pre-set Alarm Limit has been exceeded; [Icon] for suggested corrective step(s)
• ... [Icon] (W overscored); displayed output power is outside a preferred tolerance band. Increase tap for possible improvements
!Cap @display; you are attempting to adjust Setpoint above safe limit for capacitors used. Reduce Setpoint



Fault:
Heating interrupted; fault detected; correction required
[Icon] to Control zone, [Icon] Status then [Icon] for fault notice

★ AC Power; indicates HOTSHOT™ is ON; * see the Status screen for notices

Interpreting Faults, Limits and Advisories

Your HOTSHOT[®] communicates information about the way it's operating. This section explains these conditions and what you can do to respond to them.

Faults	
	Heating interrupted, fault detected; correction required to Control room, STATUS then for fault notice
	Inadequate flow to coil (FLS2, see schematic) ✓ assure water flow thru coil ✓ assure plug at PPH
	Inadequate flow to unit (FLS1, see schematic) ✓ assure water flow thru unit ✓ assure plug at CPH
	Coil shorted / increase caps
	Frequency out of range START restored after 4 sec. ✓ clean coil remove bridging ✓ call Amertherm (see caps)
Limit (auto)	
	This occurs when the unit has sensed a mis-match. It lets you know that output is being restricted due to the condition mentioned in the notice; Cutback Mode may be displayed
	Cap Over Voltage Cap Over Current Heat Sink Hot Enclosure Hot
	Adjust Cap/Cap to improve output/setpoint
	Cutback Mode CAP TEMP PROTECT REDUCE OUTPUT
	peak current protection, you note output falls with increased setpoint
	match ✓ reduce setpoint ✓ part out of coil, thru Curve? ✓ follow 'adjust taps' notices ✓ See §3.4
Limit (manual) L [®]	
	See L-asterisk on display. If enabled (§3.3), occurs if the output falls or rises outside a Lo - Hi band you specify. Use Ready output to signal your equipment
	See L-asterisk on display. Indicates output/setpoint are not well matched; more efficient configuration may be possible through output it within acceptable margins. See §3.4

You may be able to correct frequency (Adjust Caps message) by adjusting coil with more (lowers frequency) or fewer (increases frequency) turns, if your coil design is fixed, call Amertherm for replacement caps



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Multi-step profile

Case: the part is brought to temperature and held in three steps. From Control zone

Do this...

- ✓ select **Adjust Profiles**
- then

Observe...

Select Profile
▶ A B C D

- ✓ press **[A]**, this selects Profile A

End Start Time
0.0A 0.0A
Untimed▶A1 Data

Each step has 3 data points

▶100A Start A1 Data

▶100A End A1 Data

▶10s Time A1 Data

100A 50A
10s ▶A1 Data

100A 200A
20s ▶A2 Data

200A 200A
10s ▶A3 Data

This step 'activates' File A.
FileA can be (de)activated any
time by selecting
Control from: ▶Front
Panel or ▶Rear Panel

★ Ready; ★ HeatOn, note three-
stage countdown until ★ Ready
only

- ✓ with A1 selected, press **[▶]**
- ✓ use **▲/▼**, set A1 Start level
(ex: sets A1 Start = 100A)
- ✓ press **[▶]**
- ✓ use **▲/▼**, set A1 End level
(ex: sets A1 End = 10A)
- ✓ press **[▶]**
- ✓ use **▲/▼**, set A1 step Time
(ex: sets A1 Time = 10s)
- ✓ press **[▶]**
- Note A1 data points are displayed

- ✓ use **▲**, select A2
- ✓ set A2 Start = 100A
- ✓ set A2 End = 300A
- ✓ set A2 Time = 20s
- ✓ use **▲**, select A3
- ✓ set A3 Data points as shown

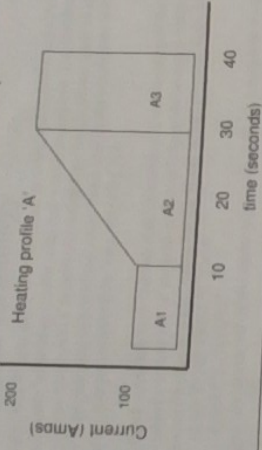
- ✓ Clean all data for indicated step

- ✓ select **System Options**, then **[▶]**
- ✓ select Control from: ▶File A
then select using **[▶]**

- ✓ **[Home]**
- ✓ press **START**

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The graph represents the 3-step heating profile in the example



- Unintimed profile steps are skipped
 - Unintimed and zero-Setpoint steps are interpreted as STOP by HOTSHOT (A4 implied, above).
- From the Home zone, Unintimed heat cycles are continuous (until STOP).



2.4 Equipment Maintenance

The effectiveness and safety of your HOTSHOT equipment can be assured for years by following these simple maintenance steps:

Frequency	Check
1 month	safety while * Ready, turn off water; assure * Fault, restore water. If no * Fault, flow switch is stuck, do not operate further! Equipment safety is compromised. Call Ameritherm
6 months	coolant level; top-off if required appearance; flush & replace if dark or cloudy system, for signs of leaks, repair
12 months	coolant dump, flush, replace coolant equipment wiring; assure all wiring is OK; replace if damaged setup; confirm ▶ RF-Setup, Caps, Taps correct
NOTE: if you use a heat transfer fluid in your equipment, dispose of spent chemicals responsibly; call an expert	

2.5 Smartburst Technology

The HOTSHOT heat stations contain one or more water cooled capacitors, rated for current under laboratory (rather than real world) conditions. The Hotshot 3.5, 5, and 7.5 kW systems make use of Smartburst technology which allows users to approach these higher current limits in actual use under many real world conditions while maintaining the safety margins required for long operating life.

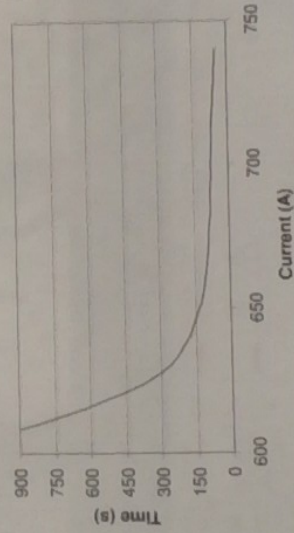
Smartburst considers both thermal time constants and self-heating in the capacitors to allow higher currents out of the heat station as long as the average value is safe, and if the continuous on time is within limits for the levels in use. Beyond these limits,

CAP TEMP PROTECT and

REDUCE OUTPUT messages are displayed on the status page, and levels are brought to a safe value.

The following chart illustrates operation for a capacitor with a laboratory current rating of 750A. When the limit becomes active, coil current is held with a ceiling of about 80% of the laboratory value (#600A).

Continuous Heat Time vs Coil RF current



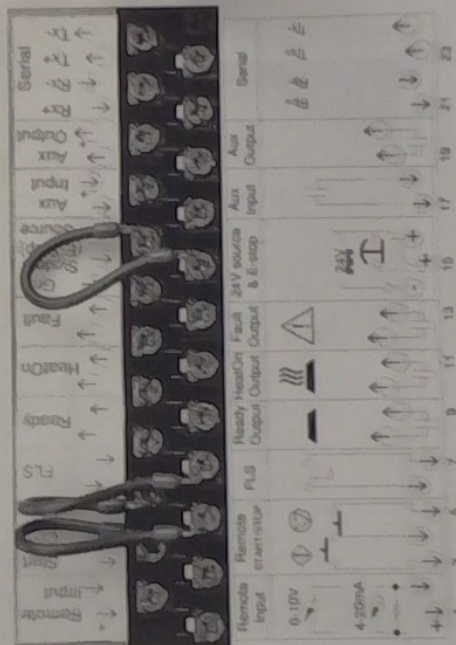
3 Customizing Your HOTSHOT

You can control your HOTSHOT™ in different ways:

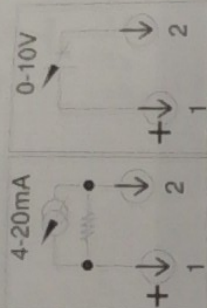
- **START / STOP** can be controlled from the front panel or remotely
- output level can be set from the front panel, remotely or with optional multi-step profiles you specify
- the state of your unit can be monitored remotely
- your load can be efficiently matched to the output with RF tap settings
- change capacitor when your coil design requires it

3.1 Rear Panel Connections

Remote operation of your HOTSHOT requires electrical connections at CTB1 and - in some cases — unit configuration from the touch-pad.



Case: Use an external signal to set the desired output level of the HOTS HOT Depending on the signal source you plan to use (0-10Vdc is the default configuration), follow these instructions to change.



Touch-and

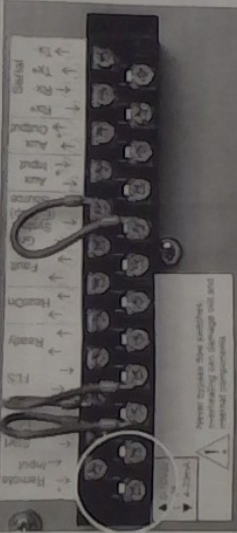
CTBI 1 = positive

(1) 2000 10000 100000

Using 4,200 A source.

CTBI 1 = positive
CTBI 2 = positive $\text{CTB}1.1 \rightarrow 2 = 250\Omega$ Observation and prediction model of d

Observe polarity with these connections!



Result: Setpoint established and held with your analog voltage or current source. To START/STOP remotely, also, see below

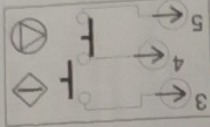


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START/STOP (CTB1:3-5)

Case: use two external signals, switches or isolated contacts to control RF heating. START signal is normally-open, active-closed, STOP is normally-closed, active-open.



2-Button... Electrical				Touch-pad	
Attach your signal				▶System Options	
CTB1:3 = START				▶Start From	
CTB1:4 = COM				▶Rear Panel	
CTB1:5 = STOP				start Type	
START:				▶2 button	
normally-open, active-closed at CTB1:3-4				to Control, to Home	
STOP:					
normally-closed, active-open at CTB1:4-5					

Result: START signal (momentary closure) initiates heating; opening STOP signal halts heating. Heating cannot commence whenever normally closed STOP signal is opened.

Case: You use one external signal, switch or isolated contact to control RF heating

1-Button... Electrical				Touch-pad	
Attach your signal				▶System Options	
CTB1:3 = START				▶Start From	
CTB1:4 = COM				▶Rear Panel	
CTB1:5 = STOP				start Type	
				▶1 button	
				to Control, to Home	

Result: Closing START initiates heating, which continues while START is active; opening START signal halts heating. Heating cannot occur whenever (jumped) STOP signal is opened.

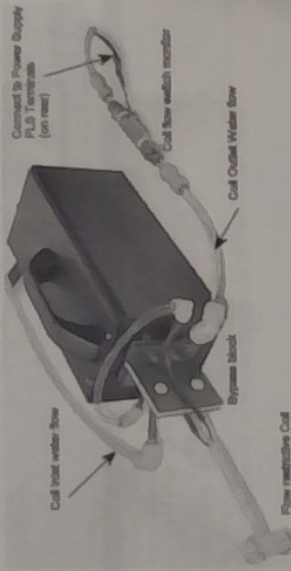
1-button Start is not recommended if you are using the liner (S2.1); the liner may re-start if button is partially depressed or is held too long.



Flow switches (CTB1-6-7)

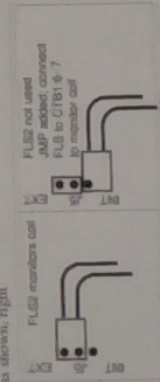
The flow switches protect your HOTSHOT from thermal damage if cooling water to the heat sink or to the coil is inadequate.

If your coil presents a restricted water path, Ameritherm may recommend the use of a "bypass-block" with your heat station. (Ask your sales representative). In this case you may be required to complete wiring for an external flow switch (sold separately).



To add your optional normally-open flow switch, remove jumper at CTB1-6-7, connect your flow switch wires. Inadequate flow to your flow switch or to either the internal flow switches will then result in a fault.

To monitor the coil with your external FLS only, add jumper and move FLS3 connection as shown, right.

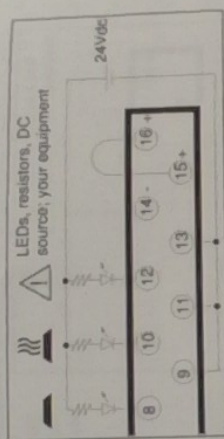




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Status signals (CTB1:8-13)

Case: You connect your 24Vdc source and monitoring equipment (ex., current-limited LEDs) to remotely track the status of the HOTSHOT

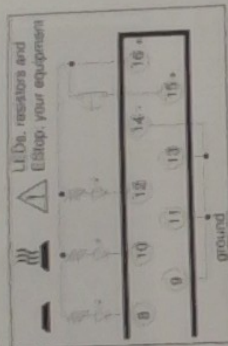


Electrical	Touch-pad
Connect to internal solid-state contacts to complete your indicator signaling (shown): CTB1:8-9 closure = ★ Ready CTB1:10-11 closure = ★ HeatOn CTB1:12-13 closure = ★ Fault No polarity to these connections. 1A limit	None

Result: Your remote 24Vdc status signals echo HOTSHOT status LEDs.

24V source & E-stop (CTB1:8-16)

Case: You use internal 24V power supply to drive your remote status signaling (ex., LEDs), you include your E-stop switch



Electrical

Connect to internal solid-state contacts to complete your indicator signaling (shown):
 CTB1:8-9 closure = • Ready
 CTB1:10-11 closure = • HeatOn
 CTB1:12-13 closure = • Fault
 No polarity to these connections. 1A limit
 Connect your N.C. E-stop (rated 24V @ 3A min. 1A, shown).

Result: Your remote 24Vdc status signals echo HOTSHOT status LEDs. E-stop trip halts RT power output and equipment operation. Reset E-stop to restart HOTSHOT at Home mode.

If no E-stop device is used, CTB1:15-16 **must be jumpered** (factory configuration) for the HOTSHOT to operate, this jumper routes all 24Vdc to the controls, display, etc.

You can combine the Ready status signal and the STOP input to halt heating if the Alarm (§3.3) is triggered. Insert jumpers per CTB1:4 = CTB1:8 and CTB1:3 = CTB1:9

Touch-pad

None

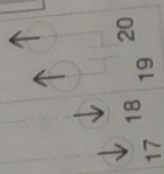


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Auxiliary Input/Output (CTRL:17-18, 19-20)

Feature not supported at this time

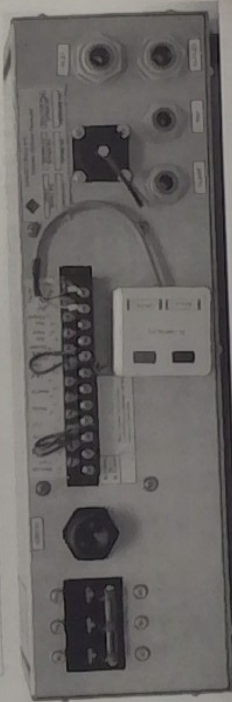
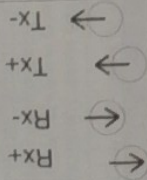


Case:

Serial Port (CTB1:21-24)

Required equipment:

- PC running Windows95 or higher/HyperTerminal
- 305-0174 Ameritherm RS485kit (or equivalent)
- PC(Tx+) → HOTSHOT(Tx+)
- PC(Tx-) → HOTSHOT(Tx-)
- PC(Rx+) → HOTSHOT(Rx+)
- PC(Rx-) → HOTSHOT(Rx-)



Why RS485?

RS485 serial communication protocol is selected for its noise immunity and its performance with long data lines. Its ability to address multiple units connected in parallel is being developed at this writing.

Purchased Separately

If you purchased your HOTSHOT including this option, the rear of your equipment will look something like this picture. The converter cable (not shown) must be connected to your computer. Configuration may require your attention.

To Configure the HOTSHOT

If purchased, we have followed these steps to prepare your HOTSHOT to make use of the serial communication option. We include these instructions for your benefit and use.

Serial communication with HOTSHOT takes place only when **Home** zone is displayed, while in **Control** mode, serial communications can be configured, but HOTSHOT ignores data on the serial port.



Case: You employ the serial port to monitor and control heat-cycle times in your process.

Do this...

✓ select **System Options**

then press **▲/▼** until...

✓ select **RS485 4 wire** to enable the serial communications feature; after this step, the port is activated, but you must take the next steps to enable serial **Start From** and **Control From**.

✓ press **▲/▼** until...

✓ once this selection has been made, press **▲** to select **Terminal Mode**

✓ select **▲** for addressing (default = 1)

✓ press **▶** to modify; (default = 38400, 0 used for broadcast, cannot be selected)

✓ select **▲** to select **baud rate**

✓ press **▶** to modify; (default = 38400, max 115,200)

✓ select **▲** to select **Start From**

✓ press **▶** to select **RS485 4 wire**

✓ select **▲** to select **Control From**

✓ press **▶** to select **RS485 4 wire**

✓ complete wiring per above guide

Observe...

Menu Selection
▶ **System Options**

Communication
▶ **RS485 4 wire**

Com Protocol
▶ **Terminal Mode**

RS485 Address
▶ **1**

RS485 Baud Rate
▶ **38400**

Start From
▶ **RS485 4 wire**

Control From
▶ **RS485 4 wire**



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To Configure Your Computer

The following is one way to configure the serial port of your computer; you may have your own devices or means of making the communications connection.

Do this...

- ✓ launch HyperTerminal from Programs > Accessories > Communication (or other serial communication application) or download from <http://www.hillgrange.com/hyper63.exe>

Observe...



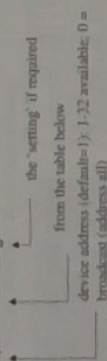
- ✓ enter a 'connection name' such as HOTSHOTComm then select:
- COM1
- 38400 baud (must match selection above)
- 8 data bits, parity = none, stop bits = 1
- flow control NONE
- ✓ click OK, save file and close HyperTerminal
- ✓ re-open HyperTerminal: open the saved file
- ✓ attach the converter kit (9-pin D-subminiature) to your COM1 port
- ✓ attach RJ-45 modular plug to lower port at rear of HOTSHOT

Configuration of your PC/converter must follow:

Data device type	SIM	Comm Mode
DCE		TX/RX/SON

Command set

address, command, argument



Parentheses indicate default/factory setting. Examples assume default address

[illegible]



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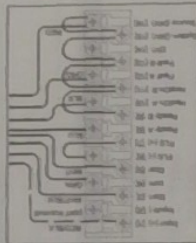
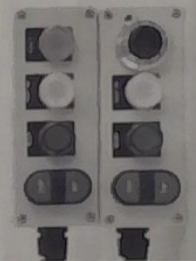
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3.2 Optional Equipment

Pendant Styles

At this writing, there are two models available:

- 3-lights, Start/Stop (upper)
- 2-lights, Start/Stop, Power Control (lower)



The 3-light model puts Stop, Start and Status indicator in your hand. The 2-light (Power Control) model lets you adjust the Setpoint from the pendant. • Fault indication is not provided at the pendant, but it is implied (• Ready and • HeatOn are both dark). • Fault status is always available at the front panel.

See 8.2.2 for Control instructions to configure the unit.

3-light model	Touch panel
Control lamp is removed from CTH	System Options Start from •Rear panel Control from •Rear panel Start type •2 button

2-light model	Touch panel
If set upper (illumination, Remote operation)	System Options Start from •Rear panel Control from •Rear panel Start type •2 button

Do this...	Observe...
Follow instruction under 8.1.4 to assemble your equipment	pendant; the one way
Attach pendant to control block to CTH	mobile handset screen, but
Secure with P-clip provided	Ready at pendant
Attach A45 spring	Ready • • HeatOn
Adjust Setpoint at front panel, one pendant to	Start/Stop handling
Adjust Setpoint at pendant; one pendant to	Ready • • HeatOn
Start/Stop handling, change Setpoint	

3) Customizing Your HOTSHOT

Footswitch

You can attach an optional footswitch to your HOTSHOT, enabling you to toggle
 • HeatOn remotely. In the first example, heating occurs only while you depress the switch; in the second, the footswitch starts a timed cycle

Continuous	Touch panel
CTB1:4→5 Jumper retained CTB1:3→4 flow switch N.O. contact	►System Options Start From ►Rear Panel Start Type ►1 button

Timer	Touch panel
CTB1:4→5 Jumper retained CTB1:3→4 flow switch N.O. contact	►System Options Start From ►Rear Panel Start Type ►2 button
✓ Adjust Timer for desired duration	
✓ Depress footswitch to start timer	



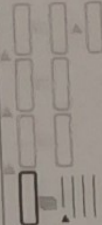
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3.3 Alarm Limits

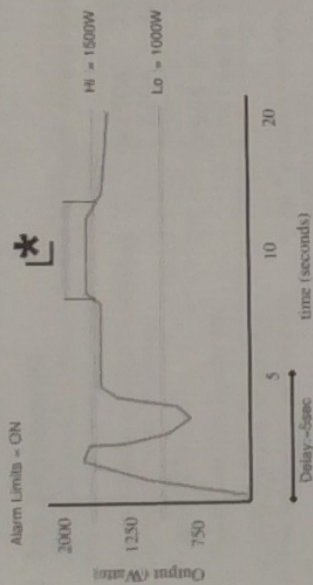
You can configure your HOTSHOT to trigger an alarm if the output level falls or rises outside a Lo - Hi band you specify after a delay period you specify. When the band has been set and the alarm enabled, the system waits the delay period after START; if the output power then violates the band, the system turns off * Ready (front panel), opens the Ready output (rear panel) and displays the L-asterisk. Heating is not interrupted, use this relay signal to switch your equipment if desired.

Alarm Limits
▶Off



Do this...	Observe...
✓ select ▶ System Options then	Alarm Limits <725W
✓ use ▲/▼, select Alarm Limits	▶On
✓ press ▶ for ON	Minimum Power
Then the display alternates between	▶ < 1000W
• minimum (<)	set to 1000 W
• maximum (>)	set to 1500 W
• delay (t)	set to 5 seconds
✓ press ▶, then ⏏ to adjust Minimum	
✓ press ▶, then ⏏ to adjust Maximum	
✓ press ▶, then ⏏ to adjust Delay	
✓ Home	
To disable the Alarms Limit feature:	
✓ select ▶ System Options then	
✓ use ▲/▼ select Alarm Limits	
✓ press ▶ until OFF is displayed	
✓ Home	

! You can combine the Ready status signal and the STOP input to halt heating if the Alarm is triggered. Insert jumpers per:
CTB1-4 = CTB1.8 and CTB1-3 = CTB1.9



In the example, the output level wanders outside the limits within the first 5 seconds, but no alarm is triggered. At the 8-second point, output exceeds the limit, triggering the limit alarm. At the 12-second point, output level again falls within the limit band and the alarm is cleared. ★ Ready is restored and the Ready output relay is closed



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3.4 Tap Adjustment

Under certain heating conditions, HOTSHOT may 'suggest' changing your RF transformer tap settings (q.3, Adviseories . . .). For example:

during ♦HeatOn, an asterisk * is shown on the display. To read the message,

✓ press for Status

There may be a message like this:

▶250A 234.00s
220A *190kHz

H565' F AM72' F
Increase Tap

- You can ignore this suggestion if your heating results are satisfactory
- the * is displayed only when there is no part in the coil



If your unit cannot meet Setpoint and a Tap change is indicated, you must adjust Tap

Tap adjustments are made with 3 wires (X, Y, G) and a jumper (Z), which are attached as needed in a 3x3 terminal block:

- X and Y deliver power from the supply
- G selects a winding scale
- Z completes winding selection

Each of these connections must be made for the work head to work properly.

Compare the position of the wires with the table to determine the TAP setting:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

K	E	F
H	D	B
J	A	Q

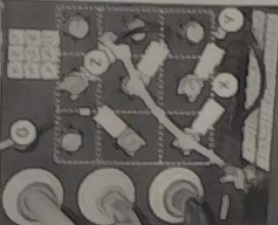


Heat stations are labeled per model; this example may differ from yours



Whenever you are unsure about the correct tap selection for your application, follow this rule: **begin with highest tap, work down**. Beginning from the highest transformer taps presents the least stress to the EASYHEAT.

Follow these steps to change taps at the work head:

Do this...	Observe...
✓ Turn off AC power switch at rear ✓ use Phillips screwdriver to remove work head cover ✓ view work head from wiring side (shown) ✓ determine present setting from table above (example = TAP 43) ✓ depending on the message, select the TAP on the chart which is numerically higher or lower than the present setting ✓ use dotted screw driver, loosen the wire(s) to be moved ✓ re-install the wire(s) as required by the table for the TAP you've selected ✓ ensure unused screws are tight ✓ replace work head cover	unit is Ready AC power LED is not lit/glowed 
✓ Navigate to Menu Selection > Heat Station ✓ select Modify Tap ✓ select the new tap setting ✓ returns to Home zone	Choose Menu ► Modify Tap until heating without *

no answer



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4 Technical Information

4.1 Mechanical

Power Supply	
Feature	Value
Dimension WxDxH	17 x 21 x 5.25 (432 x 578 x 133)
Internal access	10 exterior lock screws to open top
Mounting	10" rack or bench top
Weight	35 (15.9)
Construction	aluminum
Finish	front panel black anodized

Your system uses and produces high voltages! Only trained or guided service persons are authorized inside the equipment. Always turn off the unit and remove the AC line cord before attempting any internal service.



Remote Heat Station	
Feature	Value
Dimension HxWxD ¹	5.74x4.25x11.26 (146x126x286)
Internal access	plastic head screws
Mounting	plastic head screws
Weight (no cord)	1.12 (0.5)
Cable	10 (3.0)
Heat medium	6 (1.52)

¹ Includes mounting flange, 1/2" for handle



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4.2 Electrical

Equipment Input

Description	Model	Value	8	7.5	Units
AC voltage		230	230	230	Vrms @ 17% Jd
Current, max		15	15	15	A rms
Frequency		50 / 60	50 / 60	50 / 60	Hz
External breaker		20	20	20	A rms
Rated Power (P _{rd})		4.3	6	9	kW max
Maximum Power		4.7	6.7	10	kVA max
Power Factor		95	95	95	

Customer Input

Port	Active	CTBU	Description	Limits
Remote Input	analog	1, 2	0 - 10 Vdc (Zin = 21kΩ) 4-20mA (Zin = 250Ω)	see §3
Start	closed	3, 4	provide locked contact change ^a	
Stop	open	4, 5		
Flow Switch	closed	6, 7	opened contact = ● Flow	10Vdc @ 0.1A
Flow Stop	open	15, 16	opened contact = stop	10Vdc @ 0.1A

^a contacts rated for 24Vdc max, required wiring current 3mA

Equipment output

Description	Value	Units
Frequency	150 / 240	Hz
RL Current ¹	regulation	A rms
Power deliverable ²	3, 5, 7, 5	kW max
Eff % (see cell)	90	%

¹ and limits max based on top setting, best station model and frequency §1.4 for transformer settings

² using 7", 4-turn coil with matched calorimeter

Customer output

Port	Active	CTBU	Description	Limits
Pressure	closed	8, 9	isolated solid state output, non-polarized	24Vdc @ 1A
Heat On	closed	10, 11		
Flow	closed	12, 13		
24Vdc	closed	14, 15	ground referenced, not supported at this time	24Vdc @ 1A
Auto Port	closed	17, 18	not supported at this time	
Serial Port	closed	19, 20	not supported at this time	

4) Technical Information



4.3 Environmental

Value	Range	Units
ambient temp	40 - 105 (4 - 41)	°F (°C)
water temp	68 - 95 (20 - 35)	
flow (system minimum)	1.5 (5.7)	gpm (l/min)
pressure	40 - 80 (2.8 - 5.5)	br/g (bar)
pH	7	pH
conductivity	less than 65000	µm/cm
resistivity @ 25°C	greater than 2000	
solids	less than 150	ppm

1.) Water temperature must not fall below the dew point in any case; condensation may result in nuisance problems or damage to the equipment

2.) You must ensure that the differential (inlet minus outlet) pressure falls within this range